# AQ10.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1:**

Consider a function $f: \mathbb{R}^2 \to \mathbb{R}$ defined as:

$$f(x,y) = \begin{cases} x^2 \sin\frac{1}{x} + y & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

- **Statement 1:** $f_y$ is continuous at the origin.
- **Statement 2:** $f_x$ is continuous at the origin.

Enter the number of correct statements.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0
*1 point*

*1 point*
A function $f: \mathbb{R}^2 \to \mathbb{R}$ satisfies that $f_x(x,y) = c$ and $f_y(x,y) = k$ of all points of $\mathbb{R}^2$, where $c$ and $k$ are some constants, then which of the following option(s) is (are) true?

☐ The tangent line exists at any point of $\mathbb{R}^2$ in any direction.

☐ The tangent lines at the origin do not lie on a plane.

☐ There exist points in $\mathbb{R}^2$ for which tangent points in all directions do not exist.

☐ $f(x,y)$ can be $f(x,y) = cx + ky + \alpha$, where $\alpha$ is any real number.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
The tangent line exists at any point of $\mathbb{R}^2$ in any direction.
$f(x,y)$ can be $f(x,y) = cx + ky + \alpha$, where $\alpha$ is any real number.

*1 point*
Consider $f: \mathbb{R}^2 \to \mathbb{R}$ and suppose partial derivatives of $f(x,y)$ with respect to $x$ and $y$ are continuous at a point $(a,b)$. Then which of the following option(s) is (are) true?

☐ $f_x(a,b) = \lim\limits_{(x,y)\to(a,b)} f_x(x,y)$

☐ $f_y(a,b) = \lim\limits_{(x,y)\to(a,b)} f_y(x,y)$

☐ $f_x(a,b) \neq \lim\limits_{(x,y)\to(a,b)} f_x(x,y)$.

☐ $f_y(a,b) \neq \lim\limits_{(x,y)\to(a,b)} f_y(x,y)$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$f_x(a,b) = \lim_{(x,y)\to(a,b)} f_x(x,y)$
$f_y(a,b) = \lim_{(x,y)\to(a,b)} f_y(x,y)$

*1 point*
Consider a function $f(x,y) = x^2 + y^2$. Which of following is the tangent at the point (1,2) in the direction of (1, 1)?

- ○ $x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$
- ○ $x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$
- ○ $x = 1, y(t) = 2 + t, z(t) = 5 + \frac{6t}{\sqrt{2}}$
- ○ $x = 1 + t, y(t) = 2 + t, z(t) = 4 + \frac{6t}{\sqrt{2}}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$

**Level 2:**

*1 point*
Consider a function

$$f(x,y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0. \end{cases}.$$

Which of the following is the tangent line passing through the point (1, -1) and parallel to the line

$x(t) = \frac{t}{\sqrt{5}}, y(t) = 1 + \frac{2t}{\sqrt{5}}, z(t) = -\frac{3t}{\sqrt{5}}$x(t) = 5
√
t, y(t) = 1 + 5
√
2t, z(t) = − 5
√
3t?

○

$x(t) = 1 + t, y(t) = -1 + 2t, z(t) = -\frac{3}{\sqrt{5}}t$x(t) = 1 + t, y(t) = −1 + 2t, z(t) = − 5
√
3t

○

$x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$x(t) = 1 + t, y(t) = 1 + 2t, z(t) = −1 − 5
√
3t

○

$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$x(t) = 1 + 5
√
t, y(t) = −1 + 5
√
2t, z(t) = −1 − 5
√
3t

○

$x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$x(t) = −1 + t, y(t) = −1 + 2t, z(t) = −1 − 5
√
3t

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$x(t) = 1 + 5
√
t, y(t) = −1 + 5
√
2t, z(t) = −1 − 5
√
3t

*1 point*

Consider a function $f(x, y, z) = 2x^2 + xy + z^2$ f(x, y, z) = 2x2 + xy + z2. Which of following is the tangent at the point (1,1,1) in the direction of (1, 0, 0)

○

$(1, 1, 1, 4 + 5t)$(1, 1, 1, 4 + 5t)

○

$(1 + t, 1, 1, 4 + t)$(1 + t, 1, 1, 4 + t)

○

$(1, 1 + t, 1 + t, 4 + 5t)$(1, 1 + t, 1 + t, 4 + 5t)

○

$(1+t, 1, 1, 4+5t)$(1 + t, 1, 1, 4 + 5t)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$(1+t, 1, 1, 4+5t)$(1 + t, 1, 1, 4 + 5t)

*1 point*

Consider the following functions $f$f defined as:

$$f(x, y, z) = xyz.$$
f(x, y, z) = xyz.

Then the tangent line at the point (1,0,1) in the direction of (1,1,1) is parallel to the line

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3
√
t, y(t) = 3
√
t, z(t) = 1 + 3
√
t, u(t) = 3
√
t

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$x(t) = 1 + 3
√
t, y(t) = 3
√
t, z(t) = 1 + 3
√
t, u(t) = t 3
√

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3
√
t, y(t) = 1 + 3
√
t, z(t) = 1 + 3
√
t, u(t) = 3
√
t

○ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

$\sqrt{}$

t, y(t) = 3

$\sqrt{}$

t, z(t) = 1 + 3

$\sqrt{}$

t, u(t) = 3

$\sqrt{}$

t

Check Answers

Your score is: 0/7